MCC Entrance Test, August 2026 — Day 2 — Solutions

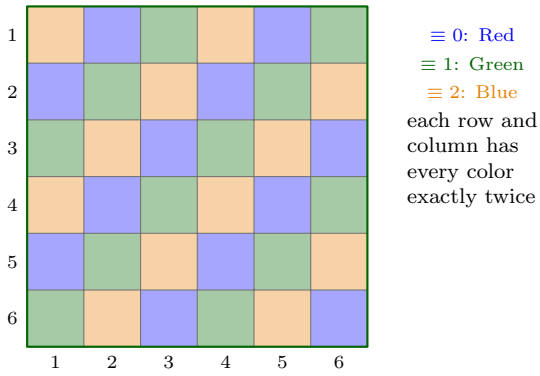
Level T3

Test Information. Level T3 — 150 minutes.
6 problems: 2 Easy (10 pts), 2 Medium (15 pts, two parts), 2 Hard (25 pts, three parts); 100 points total.
Areas: Algebra, Combinatorics (×2), Geometry, Number Theory (×2).
Rules: show your work (partial credit rewards genuine progress, but claims still need real justification at this level); no calculators; read all six before choosing where to start; box your final answers.
A past MCC entrance test, free to try: https://nghia71.github.io/readiness

Problem 1 (10 pts).[1] Every cell of an $n \times n$ grid is colored with one of three colors – Red, Green, or Blue. Find the smallest n for which, no matter how the colors are assigned, some row or some column is forced to contain three cells of the same color.

Official solution. **A coloring of a 6×6 grid with no such triple exists.**

Number the rows and columns $1, \ldots, 6$ and color the cell in row i, column j with the color determined by $(i + j) \bmod 3$ (say 0 =Red, 1 =Green, 2 =Blue). In any fixed row i, as j runs through $1, \ldots, 6$, the value $i + j \bmod 3$ cycles through $0, 1, 2$ exactly twice, so each color appears in that row exactly twice – never three times. By the symmetry of $i + j$ in i and j, the same holds for every column. So $n = 6$ admits a coloring avoiding the required triple.



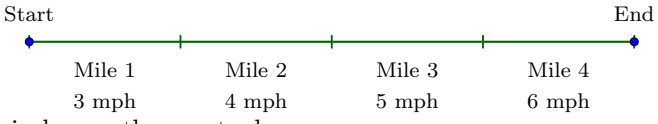
Every coloring of a 7×7 grid (or larger) contains such a triple. Take any single row of 7 cells. With only 3 possible colors, if every color appeared at most twice the row could hold at most $3 \times 2 = 6$ cells – too few for 7. So by the Pigeonhole Principle, some color appears at least 3 times in that row alone, regardless of how the grid is colored. So $n = 6$ can avoid the triple but $n = 7$ cannot, making the smallest such n equal to $\boxed{7}$. □

Problem 2 (10 pts).[2] A four-digit number $\overline{aabb}$ – whose first two digits are equal and whose last two digits are equal – is a perfect square. Find this number.

Official solution. Write $N = \overline{aabb} = 1100a + 11b = 11(100a + b)$, with $1 \le a \le 9$, $0 \le b \le 9$. Since 11 is prime, N is a perfect square only if $11 \mid (100a + b)$; as $100 \equiv 1 \pmod{11}$, this means $a + b \equiv 0 \pmod{11}$, and since $2 \le a + b \le 18$, the only possibility is $a + b = 11$.

Substituting $b = 11 - a$ gives $100a + b = 99a + 11 = 11(9a + 1)$, so $N = 121(9a + 1)$, which is a perfect square exactly when $9a + 1$ is a perfect square. Checking $a = 2, \ldots, 9$ (since $b = 11 - a \ge 0$ forces $a \le 9$, and $b \le 9$ forces $a \ge 2$), only $a = 7$ gives $9a + 1 = 64 = 8^2$. This yields $b = 4$, so $N = 121 \times 64 = \boxed{7744} = 88^2$. □

Problem 3 (15 pts).[3] A runner completes a four-mile course. She covers the first mile at 3 mph, the second mile at 4 mph, the third mile at 5 mph, and the fourth mile at 6 mph.



(a) Find the total time, in hours, the run took. [6 pts]

(b) Using your answer to part (a), find the runner's average speed for the entire course, expressed as a fraction m/n in lowest terms. Report $m + n$. [9 pts]

Official solution. **Part (a).** Each mile takes $\frac{1}{\text{speed}}$ hrs, so the total time is $\frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} = \frac{20}{60} + \frac{15}{60} + \frac{12}{60} + \frac{10}{60} = \boxed{\frac{19}{20}}$ hrs.

Part (b). Average speed is total distance divided by total time – *not* the average of the four individual speeds – so using part (a), average speed $= \frac{4}{19/20} = \frac{80}{19}$ mph. Since $\gcd(80, 19) = 1$, $m = 80$ and $n = 19$, so $m + n = \boxed{99}$. □

[1] Romanian National Mathematical Olympiad (Olimpiada Naţională de Matematică). [2] Regional Mathematical Olympiad (RMO), India. [3] Purple Comet! Math Meet, Middle School Division.

Problem 4 (15 pts).[4] Let ABC be a triangle with $\angle A = 60°$. Points K, M, N lie on sides AB, BC, CA respectively, so that $BK = KM = MN = NC$. Given that $AN = 2\,AK$,

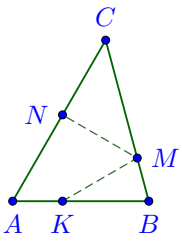
(a) Find $\angle KMN$. [4 pts]
(b) Find $\angle B$ and $\angle C$. [11 pts]

Official solution. **Part (a).**

Since $BK = KM$, $\triangle BKM$ is isosceles with apex K; as $K \in AB$, $M \in BC$, $\angle KBM = \angle ABC = \angle B$, so the base angles satisfy $\angle KMB = \angle B$.

Since $MN = NC$, $\triangle MNC$ is isosceles with apex N; as $N \in CA$, $\angle NCM = \angle ACB = \angle C$, so $\angle NMC = \angle C$ too.

Since B, M, C are collinear, $\angle BMK + \angle KMN + \angle NMC = 180°$, so $\angle KMN = 180° - \angle B - \angle C = \angle A = \boxed{60°}$.



Part (b). Since $\angle KMN = 60°$ (part a) and $KM = MN$ (given), triangle KMN is isosceles with a $60°$ apex angle, hence *equilateral*: $KN = KM = MN$, and $\angle MKN = 60°$.

Let $AK = x$, so $AN = 2x$. In triangle AKN, $\angle KAN = \angle A = 60°$, so by the Law of Cosines,

$$KN^2 = AK^2 + AN^2 - 2 \cdot AK \cdot AN \cos 60° = x^2 + 4x^2 - 2x(2x)\left(\tfrac{1}{2}\right) = 3x^2 \implies KN = x\sqrt{3}.$$

Since $AK^2 + KN^2 = x^2 + 3x^2 = 4x^2 = AN^2$, $\triangle AKN$ is right-angled at K, so $\angle AKN = 90°$.

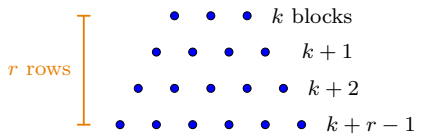
From part (a)'s isosceles triangle BKM, $\angle BKM = 180° - 2\angle B$, so $\angle AKM = 180° - \angle BKM = 2\angle B$ (as A, K, B are collinear). Since ray KN lies between rays KA and KM,

$$\angle AKM = \angle AKN + \angle NKM = 90° + 60° = 150° \implies 2\angle B = 150° \implies \angle B = \boxed{75°} \implies \angle C = \boxed{45°}.$$

□

Problem 5 (25 pts).

[5] A *stack* is an arrangement of blocks in rows, where every row (after the top one) has exactly one more block than the row directly above it. A positive integer n is called a *stack-number* if some stack uses exactly n blocks in total, with at least two rows and at least two blocks in the top row.



(a) For a stack with r rows ($r \geq 2$) and k blocks in the top row ($k \geq 2$), find a formula for the total number of blocks n, in terms of r and k. Writing $s = 2k + r - 1$, use your formula to show that $2n = r \cdot s$, and then prove that r and s are always positive integers of opposite parity, with $s > r$. [6 pts]
(b) Using part (a), show that no stack-number is ever a power of 2. [3 pts]
(c) Using parts (a)–(b), determine how many stack-numbers there are less than 100. [16 pts]

Official solution. **Part (a).** The rows hold $k, k+1, \ldots, k+r-1$ blocks, so $n = k + (k+1) + \cdots + (k+r-1) = \boxed{\dfrac{r(2k+r-1)}{2}}$. Hence $2n = r(2k+r-1)$; writing $s = 2k + r - 1$, the difference $s - r = 2k - 1$ is odd, so r and s have opposite parity. Since $k \geq 2$, $s = 2k + r - 1 \geq r + 3 > r$.

Part (b). Suppose $n = 2^m$ were a stack-number. By part (a), $2n = 2^{m+1} = r \cdot s$ with r, s of opposite parity and $2 \leq r < s$. Whichever of r, s is odd must itself divide the power of 2 equal to 2^{m+1}, which forces that odd factor to equal 1 – but the odd factor is either $r \geq 2$ or $s > r \geq 2$, both > 1. This is a contradiction, so $\boxed{\text{no power of 2 is a stack-number}}$.

[4] Iranian Geometry Olympiad (IGO). [5] New Zealand Mathematical Olympiad (NZMO).

Part (c). Plan: count the values $n < 100$ that *fail* to be stack-numbers, then subtract from 99.

Step 1 – every consecutive-sum comes from an odd divisor. Extend part (a)'s algebra to also allow $r = 1$ (the trivial single-row stack, i.e. writing n as itself) and $k \geq 1$; part (a)'s relation $s = 2k + r - 1$ still holds, just solved for $k = \frac{s-r+1}{2}$ instead.

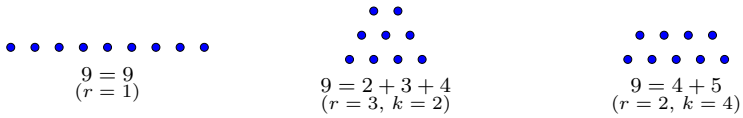
So every factorization $2n = r \cdot s$ with $r < s$ of opposite parity still describes a way of writing n as r consecutive positive integers starting at that k.

These factorizations match up perfectly with the *odd* divisors of n. Given an odd divisor d of n: since d divides n, the quotient n/d is a whole number, so $2n/d = 2(n/d)$ is even; pairing d with $2n/d$ (letting r be whichever of the two is smaller and s the larger) gives exactly one opposite-parity factorization of $2n$.

Conversely, in any opposite-parity factor pair of $2n$, the odd member must itself divide n (it divides $2n$ without using $2n$'s factor of 2) – so every factorization arises this way, from some odd divisor of n.

For example, $n = 9$ *has odd divisors* $1, 3, 9$*, giving exactly three representations:*

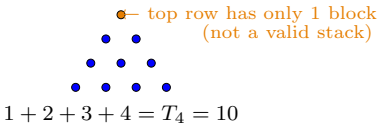
$$9 = 9 \ (r{=}1), \qquad 9 = 2 + 3 + 4 \ (r{=}3,\ k{=}2), \qquad 9 = 4 + 5 \ (r{=}2,\ k{=}4).$$



So *the number of ways to write* n *as* $r \geq 1$ *consecutive positive integers equals the number of odd divisors of* n, and dropping the trivial $r = 1$ case, *the number of ways with* $r \geq 2$ *equals (odd divisors of* n*)* -1.

Step 2 – the stack's extra rule.

A real stack also needs $k \geq 2$, since the top row can't hold just 1 block – so any representation with $k = 1$ isn't an actual stack and must be excluded. With $k = 1$ in Step 1's formula gives the "staircase from 1": $n = 1 + 2 + \cdots + r = \frac{r(r+1)}{2} = T_r$, the rth triangular number.



Since $T_1 < T_2 < T_3 < \cdots$ is strictly increasing, at most one value of r can satisfy $T_r = n$ – so n has a $k = 1$ representation for at most one r, and only when n itself is triangular.

Step 3 – when n is not a stack-number. Combining Steps 1–2, n fails to be a stack-number exactly when either:

- it has 0 representations with $r \geq 2$ – i.e. only 1 odd divisor, i.e. n is a power of 2 (this is part (b)); or
- it has exactly 1 representation with $r \geq 2$ – i.e. exactly 2 odd divisors, i.e. its odd part is prime – *and* that lone representation is the excluded $k = 1$ case, i.e. n is triangular.

Step 4 – count the exclusions below 100.

Powers of 2*:* $1, 2, 4, 8, 16, 32, 64$ – 7 values.

Triangular numbers with a prime odd part: checking all 13 triangular numbers below 100,

r	1	2	3	4	5	6	7	8	9	10	11	12	13
T_r	1	3	6	10	15	21	28	36	45	55	66	78	91
odd part	1	3	3	5	15	21	7	9	45	55	33	39	91
prime?	pow. 2	✓	✓	✓			✓						

$T_1 = 1$ is a power of 2, already counted above. Reading off the "prime?" row, the odd part is prime for $T_2 = 3$, $T_3 = 6$, $T_4 = 10$, and $T_7 = 28$ – these 4 values are newly excluded.

Every other triangular number's odd part is composite (e.g. T_8's is $9 = 3^2$, T_9's is $45 = 3^2 \cdot 5$), which by Step 3 means each has a second, valid representation with $k \geq 2$ – so those n remain stack-numbers.

Step 5 – total. With no overlap beyond T_1 (already handled), $7 + 4 = 11$ values in $\{1, \ldots, 99\}$ fail to be stack-numbers, so the number of stack-numbers less than 100 is $99 - 11 = \boxed{88}$. □

Problem 6 (25 pts).[6] A positive integer m satisfies $m^{1003} \equiv 19 \pmod{77}$.

(a) Find the remainder when m is divided by 11. [12 pts]
(b) Find the remainder when m is divided by 7. [3 pts]
(c) Using parts (a)–(b), find the remainder when m is divided by 77. [10 pts]

Lemma (Cubing Is Injective Mod 11)
If $x^3 \equiv y^3 \pmod{11}$ for some $x, y \not\equiv 0 \pmod{11}$, then $x \equiv y \pmod{11}$ – i.e. distinct nonzero residues mod 11 always have distinct cubes.

Proof. Let z be the residue with $z \equiv x \cdot y^{-1} \pmod{11}$ (an inverse for y exists since 11 is prime), so $z^3 \equiv 1 \pmod{11}$. Since $3 \times 7 = 21 = 2 \times 10 + 1$, raising both sides to the 7th power gives

$$z^{21} = (z^3)^7 \equiv 1^7 = 1 \pmod{11},$$

while writing the same power the other way round, Fermat's Little Theorem gives $z^{21} = (z^{10})^2 \cdot z \equiv 1^2 \cdot z = z \pmod{11}$. So $z \equiv 1 \pmod{11}$, i.e. $x \equiv y \pmod{11}$. □

Official solution. **Part (a).** Since $19 \not\equiv 0 \pmod{11}$, $11 \nmid m$, so by Fermat's Little Theorem $m^{10} \equiv 1 \pmod{11}$. Since $1003 = 100 \cdot 10 + 3$, $m^{1003} \equiv m^3 \pmod{11}$, so the condition becomes $m^3 \equiv 19 \equiv 8 \pmod{11}$. By the Lemma above, this congruence has *at most one* solution m mod 11; checking small values, $2^3 = 8$, so $m \equiv \boxed{2} \pmod{11}$.

Part (b). Since $19 \not\equiv 0 \pmod{7}$, $7 \nmid m$, so by Fermat's Little Theorem $m^6 \equiv 1 \pmod{7}$. Since $1003 = 167 \cdot 6 + 1$, $m^{1003} \equiv m \pmod{7}$, so $m \equiv 19 \equiv \boxed{5} \pmod{7}$.

Part (c). By the Chinese Remainder Theorem, the system

$$\begin{cases} m \equiv 2 \pmod{11} \\ m \equiv 5 \pmod{7} \end{cases}$$

from parts (a)–(b) has a unique solution mod 77. To find it, write $m = 11j + 2$; substituting into the second congruence, $11j + 2 \equiv 5 \pmod{7}$, thus $4j \equiv 3 \pmod{7}$. Since $4 \times 2 = 8 \equiv 1 \pmod{7}$, $4^{-1} \equiv 2 \pmod{7}$, so $j \equiv 3 \times 2 = 6 \pmod{7}$. Writing $j = 7t + 6$,

$$m = 11(7t + 6) + 2 = 77t + 68.$$

So the remainder when m is divided by 77 is $\boxed{68}$ (and indeed $68 \equiv 2 \pmod{11}$, $68 \equiv 5 \pmod{7}$, confirming both parts above). □

[6] Korean Mathematical Olympiad (KMO), First Round.